

# William Andrews

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## EDUCATION

**The College of William & Mary - Computer Science GPA: 3.92**

*B.S. in Computer Science and Mathematics*

Williamsburg, VA

*Sept. 2023 – Expected May 2027*

## WORK EXPERIENCE

**Navigation Systems & Algorithms – Software Engineer Intern**

May 2026 – Present

*Xona Space Systems*

*Burlingame, CA*

- Improving **C++ algorithms** related to: **satellite navigation**, **orbital mechanics**, **signal processing**, as well as **signal encryption** and **authentication**.
- Strengthening existing C++ testing frameworks for the math in **Pulsar**: the most **advanced** and **resilient** positioning, navigation, and timing infrastructure in the world.
- Performing in-depth **performance analyses** and **increasing systems-level performance** of existing signal processing algorithms for Pulsar.
- Improved performance of C++ orbital mechanics algorithms, as well as software defined radio tracking and pulse blanking algorithms applied to a watermarking procedure.

## RELEVANT COURSEWORK — GRADUATE CLASSES IN **Bold**

- Top Four Fall 2026: Numerical Analysis – CS Honors Thesis – **Advanced Operating Systems** – Nonlinear Dynamics
- Compiler Optimization for High Perf. Computing** – Operating Systems – **Advanced Computer Architecture**
- Systems Programming** – Directed Study in High Performance Computing – Differential Equations
- Honors Real Analysis – Graph Theory – Intermediate Linear Algebra – Abstract Algebra – Probability

## RESEARCH & PROJECTS

**Honors Thesis: Entropy-based Perf. Optimization for World Action Models**

Feb. 2026 – Present

*Advisor: Prof. Bin Ren, Department of Computer Science*

*Williamsburg, VA*

- Researching World Action Models and entropy-based methods to optimize computation time and memory management.
- Analyzing **KV-cache bottlenecks** in related World Action Model and Vision Language Action Model frameworks, particularly regarding data locality and memory constraints during inference.
- Exploring **entropy-based token filtering and pruning methods** and related optimizations for KV-cache compression in long context windows.
- Developed from scratch a basic C++ LLM inference engine: **llm.cpp**, to examine LLM forward passes under-the-hood.
- Year long research will culminate in an **undergraduate honors thesis** and **potential publication**.

**Directed Study in High Performance Computing**

Aug. 2025 – Dec. 2025

*Advisor: Prof. Robert Michael Lewis, Department of Computer Science*

*Williamsburg, VA*

- Researched and implemented high-performance CPU/GPU algorithms with **C/C++**, **OpenMP**, and **MPI**, focused on numerical methods, memory/cache efficiency, and multi-threaded patterns.
- Implemented **SIMD** vectorized matrix operations in C using **AVX-512**; benchmarked against auto-vectorization; improving code for CPU data pipelining to increase cache and bandwidth performance.
- Final Project:** Developed a high-performance 2D Jacobi and Gauss–Seidel solver in C++ for **Laplace Partial Differential Equations**, implemented iterative stencils and parallelization to efficiently compute steady-state solutions on large grids.

**Graph Theoretic Transit Routing Engine**

Aug. 2025

*Personal Project*

*Alexandria, VA*

- Created a Waze-inspired **routing engine** to explore open source shortest path computation methods.
- Invented a multi-threaded, bidirectional A\*-based algorithm, reducing long distance computation times.

**Matroid Algorithm Optimization**

May 2025 – Aug. 2025

*Personal Project*

*Alexandria, VA*

- Designed a **C++ framework** to solve matroids in **combinatorial optimization**.
- Implemented matroid greedy algorithms to find minimum spanning trees in regular and bipartite graphs, find the minimum basis for matrices, and solve abstract set systems.

## SKILLS & TOOLS

**Languages:** C, C++, Python, and Go.

**Topics:** High Perf. Computing, Low-Level C++ Optimization, Diff. Equations, Linear Algebra, Graph Theory, Real Analysis.